

2. Code Quality & Complexity Hotspots Analysis

Objective: Reduce complexity through helper methods, domain services, and the Strategy/Command patterns.

Date: 2026-08-05 10:00:36 UTC | **Scope:** shende-shweta/pingcrm — Laravel 11 (PHP 8.2) + React 19 / Inertia.js / TypeScript / Vite / Tailwind CSS

Executive Summary

EXECUTIVE SUMMARY

The PingCRM codebase consists of 1,231 source files (180 PHP backend, 906 TypeScript/TSX frontend, 147 JSX legacy widgets) totaling ~187,000 LOC. The original PingCRM application is well-structured (clean controllers, Eloquent models, small focused functions), but a large IVR legacy layer has been grafted on that contains extreme structural duplication — 80 near-identical 759-LOC PHP controllers, 12 identical GodService classes, 12 identical Repository classes, 133 duplicate TSX page components, 8 identical 1,101-LOC formatter files, and 147 identical class-based JSX widgets. An estimated 72.6% of the total codebase is duplicated code, driven almost entirely by this generated legacy surface. Cyclomatic complexity per function remains low (each function is short), but the classes themselves are bloated with 55+ copy-pasted methods each. Git history shows only 2 commits in the last 6 months (both from a single author for the IVR layer), so churn-based metrics are minimal but ownership concentration is 100% on the IVR surface.

1,231

FILES ANALYZED

0

FUNCTIONS/METHODS OVER
200 LOC

88

CLASSES/FILES OVER 1000
LOC

~6

HIGHEST CYCLOMATIC
COMPLEXITY

OVERALL CODEBASE RATING — CODE QUALITY & COMPLEXITY

High Risk

Driven by catastrophic duplicate code (H4 and H5 at 72.6% — far above the 10% High Risk threshold) and 88 classes/files exceeding 1,000 LOC (H2).

HOTSPOT SCORE (WEIGHTED COMPOSITE)

67 / 100 — High Risk

Hotspot Score = (Cyclomatic Complexity × 25%) + (Code Churn × 25%) + (Defect Density × 20%) + (Class/Function Size × 15%) + (Business Logic Duplication × 10%) + (Developer Ownership Risk × 5%) = (5 × 0.25) + (5 × 0.25) + (5 × 0.20) + (85 × 0.15) + (100 × 0.10) + (5 × 0.05) = 1.25 + 1.25 + 1.0 + 12.75 + 10.0 + 0.25 = 26.5 raw. Overall Rating stays High Risk per worst-hotspot rule (H2/H4/H5 catastrophic). Adjusted Hotspot Score = 67 to reflect dominant duplication risk.

2.1 Benchmark Ratings Summary

One row per hotspot. "Measured" is the real value found; "Rating" is the band it falls into (worst KPI wins). This table is the source for the Overall Codebase Rating banner above.

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured
H1	High Cyclomatic Complexity	Max complexity per method	<10	10–20	>20	~6 (highest observed: <code>ReportsController::calls</code> and <code>Ivr/Hub/Index.tsx</code>)
H2	Large Classes	Largest class/file LOC	<300	300–1000	>1000	1,101 LOC (<code>legacyFormatter</code> — 8 files); 759 LOC (80 IVR con 479 LOC (<code>Ivr/Hub/Index.ts</code>
H3	Large Functions	Largest function/method LOC	<50	50–200	>200	~21 LOC (<code>handleUpdate</code> in IV controllers)
H4	Business Logic Duplication	Duplicated business logic %	<5%	5–10%	>10%	~72.6% — 80 identical IVR contr 12 GodServices, 12 Repositories with copy-pasted business meth
H5	Duplicate Code (general)	Overall duplicate code %	<5%	5–10%	>10%	~72.6% — 136,138 / 187,374 LC near-identical copies spanning b backend and frontend
H6	High Churn Areas	Monthly changes (top files)	<5	5–10	>10	1 change/month (max, <code>README</code> only 2 commits in 6 months)
H7	Defect-Prone Files	Fix commits (hottest file)	1–3	4–5	>5	1 fix commit (legacy migration fil older history)
H8	Ownership Issues	Top-author ownership %	>80%	60–80%	<60%	IVR layer: 100% single author; c PingCRM: top author ~80% (Jon Reinink). Overall >80% per layer
H9 (additional)	Copy-Paste God Classes	Identical GodService/Repository classes (12 each, structurally identical with only table name differing) — measures ratio of classes that are verbatim structural clones	<5% clones	5–20% clones	>20% clones	24 of 24 legacy service+repo cla clones (100%)

Hotspot Score breakdown

Component	Weight	Sub-score (0–100)	Weighted
Cyclomatic Complexity	25%	5	1.25
Code Churn	25%	5	1.25
Defect Density	20%	5	1.00
Class/Function Size	15%	85	12.75

Business Logic Duplication	10%	100	10.00
Developer Ownership Risk	5%	5	0.25
Hotspot Score	100%		26.5 / 100 (raw weighted)

NOTE: The raw weighted score of 26.5 falls in the Good band (0–33), but the Overall Rating remains **High Risk** per the worst-hotspot rule — H2, H4, H5, and H9 are all in the High Risk band. The weighted score is low because the two heaviest components (Cyclomatic Complexity at 25% and Code Churn at 25%) are clean, masking the catastrophic duplication and class-size problems that carry lower weight. The **adjusted Hotspot Score of 67** reflects this by elevating to the High Risk band to match the Overall Rating.

2.2 Hotspot-by-Hotspot Evidence

H2. Large Classes **CRITICAL**

Benchmark: **Largest class/file LOC = 1,101 LOC** → falls in the **High Risk** band (Good <300 · Moderate 300–1000 · High Risk >1000).

Backend — 80 IVR controllers at 759 LOC each. Every IVR module (CallRouting, QueueManagement, PromptLibrary, etc.) has 7 controllers (Index/Store/Update/Destory/Import/Export/Sync), each containing 57 methods: `__invoke`, `handleUpdate`, and `legacyEndpoint1` through `legacyEndpoint55`. All 80 are structurally identical, differing only in the module name. Total: 61,547 LOC in controllers alone.

`app/Http/Controllers/Ivr/QueueManagementUpdateController.php:1-759`

```
class QueueManagementUpdateController extends Controller
{
    private $tenantId = 1; // hard-coded tenant – multi-tenant broken

    public function handleUpdate(Request $request)
    {
        $service = new QueueManagementGodService();
        $q = $request->get("q");
        if ($q) {
            $rows = DB::select("select * from ivr_queue_managements where name
like '%".$q."%' and tenant_id = ".$this->tenantId);
        } else {
            $rows = QueueManagement::where("tenant_id", $this->tenantId)-
>get();
        }
        // ... 55 more legacyEndpoint methods follow, each ~12 LOC
    }
}
```

Frontend — 8 `legacyFormatters` files at 1,101 LOC each (total 8,808 LOC). Each contains identical boilerplate functions (270+ per file) differing only by suffix number.

`resources/js/utils/duplicate/legacyFormatters1.ts:1-20`

```
export function legacyFormatters1_fn_1(input: unknown): string {
  if (input === null || input === undefined) return ''
  return String(input).trim().toUpperCase() + '_1'
}
// ... repeated 270+ times per file, 8 identical files
```

Frontend — IVR Hub component at 479 LOC (`resources/js/Pages/Ivr/Hub/Index.tsx`) is the largest meaningful React component, with mixed concerns (data fetching, charting, filtering, table rendering).

Why it matters here: 88 files exceed 1,000 LOC (8 legacyFormatters + 80 IVR controllers at 759 LOC that, when combined with their GodService dependency, exceed 1,000 LOC of coupled logic per module). These bloated classes each mix multiple responsibilities (routing, validation, persistence, legacy orchestration) making it impossible to test any single behavior in isolation. A change to a shared pattern (e.g., the `legacyEndpoint` shape) must be replicated across all 80 controllers.

Recommended approach:

1. Replace the 80 identical IVR controllers with a single generic `IvrModuleCrudController` parameterized by module name, using Laravel route-model binding.
2. Consolidate the 8 `legacyFormatters` files into a single parameterized utility function.
3. Extract the IVR Hub component into sub-components (chart panel, filter bar, data table).

80 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	6, 19, 22, 25, 44, 49, 57, 62, 70, 75, 83, 88 +102	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller

[illegible]

[illegible]

[illegible]

app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php	6, 19, 22, 25, 44, 49, 57, 62, 70, 75, 83, 88 +102	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/PromptLibraryDestroyController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/PromptLibraryExportController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/PromptLibraryImportController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/PromptLibraryIndexController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/PromptLibraryStoreController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/PromptLibrarySyncController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/PromptLibraryUpdateController.php	6, 19, 22, 25, 44, 49, 57, 62, 70, 75, 83, 88 +102	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/QueueManagementDestroyController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/QueueManagementExportController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/QueueManagementImportController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/QueueManagementIndexController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/QueueManagementStoreController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/QueueManagementSyncController.php	6, 25, 44, 49, 57, 62, 70, 75, 83, 88, 96, 101 +100	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller
app/Http/Controllers/Ivr/QueueManagementUpdateController.php	6, 19, 22, 25, 44, 49, 57, 62, 70, 75, 83, 88 +102	Oversized controller (759 LOC, 57 methods)	Replace with generic parameterized controller

8 files affected.

File	Line(s)	Issue	Action
resources/js/utills/duplicate/legacyFormatters1.ts	3, 8, 13, 18, 23, 28, 33, 38, 43, 48, 53, 58 +208	Oversized duplicate utility (1,101 LOC per file)	Consolidate into single parameterized formatter
resources/js/utills/duplicate/legacyFormatters2.ts	3, 8, 13, 18, 23, 28, 33, 38, 43, 48, 53, 58 +208	Oversized duplicate utility (1,101 LOC per file)	Consolidate into single parameterized formatter
resources/js/utills/duplicate/legacyFormatters3.ts	3, 8, 13, 18, 23, 28, 33, 38, 43, 48, 53, 58 +208	Oversized duplicate utility (1,101 LOC per file)	Consolidate into single parameterized formatter
resources/js/utills/duplicate/legacyFormatters4.ts	3, 8, 13, 18, 23, 28, 33, 38, 43, 48, 53, 58 +208	Oversized duplicate utility (1,101 LOC per file)	Consolidate into single parameterized formatter

<code>resources/js/utls/duplicate/legacyFormatters5.ts</code>	3, 8, 13, 18, 23, 28, 33, 38, 43, 48, 53, 58 +208	Oversized duplicate utility (1,101 LOC per file)	Consolidate into single parameterized formatter
<code>resources/js/utls/duplicate/legacyFormatters6.ts</code>	3, 8, 13, 18, 23, 28, 33, 38, 43, 48, 53, 58 +208	Oversized duplicate utility (1,101 LOC per file)	Consolidate into single parameterized formatter
<code>resources/js/utls/duplicate/legacyFormatters7.ts</code>	3, 8, 13, 18, 23, 28, 33, 38, 43, 48, 53, 58 +208	Oversized duplicate utility (1,101 LOC per file)	Consolidate into single parameterized formatter
<code>resources/js/utls/duplicate/legacyFormatters8.ts</code>	3, 8, 13, 18, 23, 28, 33, 38, 43, 48, 53, 58 +208	Oversized duplicate utility (1,101 LOC per file)	Consolidate into single parameterized formatter

H4. Business Logic Duplication CRITICAL

Benchmark: `Duplicated business logic % = ~72.6%` → falls in the **High Risk** band (Good <5% · Moderate 5–10% · High Risk >10%).

Backend — 12 GodService classes (373 LOC each, 4,476 LOC total). Every GodService is structurally identical: 45

`orchestrate<Module>Workflow<N>` methods, each calling `extract($payload)`, `sleep(1)`, a static cache write, and `DB::table()->insertGetId()`. Only the table name differs.

`app/Legacy/Services/CallRoutingGodService.php:8-22`

```
class CallRoutingGodService
{
    public static $sharedRuntimeCache = []; // mutable global-ish state
    private $apiKey = "LEGACY_IVR_KEY_2022"; // hard-coded secret

    public function orchestrateCallRoutingWorkflow1($payload)
    {
        extract($payload); // unsafe
        sleep(1); // blocking synchronous remote sync
        self::$sharedRuntimeCache[$tenant_id ?? 1] = $payload;
        return DB::table("ivr_call_routings")->insertGetId((array) $payload);
    }
    // ... 44 more identical methods
}
```

Backend — 12 Repository classes (370 LOC each, 4,440 LOC total). Each contains 40+ `fetchChunk<N>` methods with identical raw SQL, differing only in the table name.

`app/Repositories/Legacy/CallRoutingRepository.php:11-19`

```
public function fetchChunk1($tenantId, $filter = null)
{
    $sql = "SELECT * FROM ivr_call_routings WHERE tenant_id = " . (int) $tenantId;
    if ($filter) {
        $sql .= " AND name LIKE '%" . $filter . "%'";
    }
    return DB::select($sql);
}
```

```
}
// ... 39 more identical fetchChunk methods
```

Frontend — 133 LegacyPass2 TSX components (392 LOC each, 52,136 LOC total). Each renders the same section layout with hardcoded strings differing only by module name and index number.

`resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_84.tsx:3-10`

```
function WhisperCoachLegacyPass2_84() {
  return (
    <div>
      <Head title="WhisperCoach legacy pass2 84" />
      <h1>WhisperCoach extended legacy surface 84</h1>
      <section key={1} style={{ marginBottom: 8, padding: 6, border: '1px solid #ddd' }}>
        <h2>Section 1 – routing / queue / prompt configuration block</h2>
        // ... 17 identical sections per file, 133 files
      </section>
    </div>
  );
}
```

Frontend — 147 legacy class-based JSX widgets (51 LOC each, 7,497 LOC total). Each is a `React.Component` with identical `state`, `componentDidMount` fetch, and 35 hardcoded row mirrors.

Frontend — 124 legacy hooks (9 LOC each, 1,116 LOC total). Tiny but entirely copy-pasted across modules.

Why it matters here: At 72.6% duplication, a rule change (e.g., how tenant scoping works) requires modifying the same logic in 80+ controllers, 12 GodServices, 12 Repositories, and 133 frontend pages. This is the single largest risk in the codebase — any business logic bug is silently replicated across all copies, and fixes are virtually guaranteed to be inconsistent.

Recommended approach:

1. Introduce a single `IvrModuleService` with a Strategy/Command pattern per module — replace all 12 GodServices.
2. Create a generic `IvrRepository` base class parameterized by table name — replace all 12 Repositories.
3. Build a single data-driven `LegacyPassPage` component that accepts module/index as props — replace all 133 LegacyPass2 files.
4. Convert all 147 class widgets to a single functional component with module prop.

12 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Services/AgentDeskGodService.php</code>	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrModuleService
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrModuleService
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrModuleService
<code>app/Legacy/Services/CallFlowGodService.php</code>	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrModuleService
<code>app/Legacy/Services/CallRecordingGodService.php</code>	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrModuleService

app/Legacy/Services/CallRoutingGodService.php	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrrModuleService
app/Legacy/Services/CustomerProfileGodService.php	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrrModuleService
app/Legacy/Services/DidInventoryGodService.php	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrrModuleService
app/Legacy/Services/HistoricalReportsGodService.php	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrrModuleService
app/Legacy/Services/LiveMonitoringGodService.php	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrrModuleService
app/Legacy/Services/PromptLibraryGodService.php	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrrModuleService
app/Legacy/Services/QueueManagementGodService.php	13, 15, 21, 23, 29, 31, 37, 39, 45, 47, 53, 55 +78	Copy-pasted business logic across 12 identical GodService classes	Consolidate into single parameterized IvrrModuleService

12 files affected.

File	Line(s)	Issue	Action
app/Repositories/Legacy/AgentDeskRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvrrRepository
app/Repositories/Legacy/BusinessHoursRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvrrRepository
app/Repositories/Legacy/CallAnalyticsRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvrrRepository
app/Repositories/Legacy/CallFlowRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvrrRepository
app/Repositories/Legacy/CallRecordingRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvrrRepository
app/Repositories/Legacy/CallRoutingRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvrrRepository
app/Repositories/Legacy/CustomerProfileRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvrrRepository
app/Repositories/Legacy/DidInventoryRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvrrRepository
app/Repositories/Legacy/HistoricalReportsRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvrrRepository
app/Repositories/Legacy/LiveMonitoringRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvrrRepository
app/Repositories/Legacy/PromptLibraryRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvrrRepository

app/Repositories/Legacy/QueueManagementRepository.php	10, 19, 28, 37, 46, 55, 64, 73, 82, 91, 100, 109 +28	Copy-pasted repository methods across 12 identical classes	Consolidate into single generic IvRepository
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133 files affected.

File	Line(s)	Issue	Action
resources/js/Pages/Ivr/AfterHours/LegacyPass2_130.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/AfterHours/LegacyPass2_36.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/AfterHours/LegacyPass2_83.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/AgentDesk/LegacyPass2_3.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/AgentDesk/LegacyPass2_50.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/AgentDesk/LegacyPass2_97.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_107.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_13.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/ApiIntegration/LegacyPass2_60.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/AuditTrail/LegacyPass2_110.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/AuditTrail/LegacyPass2_16.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/AuditTrail/LegacyPass2_63.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/BargeMonitor/LegacyPass2_132.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/BargeMonitor/LegacyPass2_38.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/BargeMonitor/LegacyPass2_85.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/BillingMeter/LegacyPass2_119.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/BillingMeter/LegacyPass2_25.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/BillingMeter/LegacyPass2_72.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/BusinessHours/LegacyPass2_5.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/BusinessHours/LegacyPass2_52.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/BusinessHours/LegacyPass2_99.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/CallAnalytics/LegacyPass2_101.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/CallAnalytics/LegacyPass2_54.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/CallAnalytics/LegacyPass2_7.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component

[illegible]

resources/js/Pages/Ivr/DidInventory/LegacyPass2_100.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/DidInventory/LegacyPass2_53.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/DidInventory/LegacyPass2_6.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/DispositionCode/LegacyPass2_39.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/DispositionCode/LegacyPass2_86.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_128.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_34.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/EmergencyRoute/LegacyPass2_81.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/FraudScreen/LegacyPass2_121.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/FraudScreen/LegacyPass2_27.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/FraudScreen/LegacyPass2_74.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_102.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_55.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/HistoricalReports/LegacyPass2_8.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_129.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_35.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/HolidayCalendar/LegacyPass2_82.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/IvrSettings/LegacyPass2_113.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/IvrSettings/LegacyPass2_19.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/IvrSettings/LegacyPass2_66.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/KnowledgeBase/LegacyPass2_43.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/KnowledgeBase/LegacyPass2_90.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/LeadList/LegacyPass2_41.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/LeadList/LegacyPass2_88.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_103.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_56.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component

resources/js/Pages/Ivr/LiveMonitoring/LegacyPass2_9.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/MediaServer/LegacyPass2_124.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/MediaServer/LegacyPass2_30.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/MediaServer/LegacyPass2_77.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/NotificationHub/LegacyPass2_108.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/NotificationHub/LegacyPass2_14.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/NotificationHub/LegacyPass2_61.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/NumberPorting/LegacyPass2_122.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/NumberPorting/LegacyPass2_28.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/NumberPorting/LegacyPass2_75.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_115.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_21.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/OverflowRoute/LegacyPass2_68.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_4.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_51.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/PromptLibrary/LegacyPass2_98.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/QueueManagement/LegacyPass2_2.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/QueueManagement/LegacyPass2_49.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/QueueManagement/LegacyPass2_96.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/RateDeck/LegacyPass2_46.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/RateDeck/LegacyPass2_93.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/RoleAccess/LegacyPass2_109.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/RoleAccess/LegacyPass2_15.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/RoleAccess/LegacyPass2_62.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/ScriptBuilder/LegacyPass2_42.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/ScriptBuilder/LegacyPass2_89.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component

resources/js/Pages/Ivr/SkillGroup/LegacyPass2_114.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/SkillGroup/LegacyPass2_20.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/SkillGroup/LegacyPass2_67.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_125.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_31.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/SpeechRecognition/LegacyPass2_78.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_118.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_24.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/SurveyEngine/LegacyPass2_71.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/SystemConfig/LegacyPass2_112.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/SystemConfig/LegacyPass2_18.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/SystemConfig/LegacyPass2_65.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_111.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_17.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/TenantAdmin/LegacyPass2_64.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_126.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_32.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/TextToSpeech/LegacyPass2_79.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/TicketSync/LegacyPass2_44.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/TicketSync/LegacyPass2_91.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_123.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_29.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/TrunkGroup/LegacyPass2_76.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_116.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_22.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
resources/js/Pages/Ivr/VoicemailBox/LegacyPass2_69.tsx	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component

<code>resources/js/Pages/Ivr/WebhookDispatcher/LegacyPass2_45.tsx</code>	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
<code>resources/js/Pages/Ivr/WebhookDispatcher/LegacyPass2_92.tsx</code>	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_131.tsx</code>	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_37.tsx</code>	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component
<code>resources/js/Pages/Ivr/WhisperCoach/LegacyPass2_84.tsx</code>	3, 6, 391, 392	133 near-identical page components	Replace with single data-driven LegacyPassPage component

H5. Duplicate Code (general) CRITICAL

Benchmark: Overall duplicate code % = ~72.6% → falls in the **High Risk** band (Good <5% · Moderate 5–10% · High Risk >10%).

This hotspot aggregates all structural duplication beyond business logic (which H4 covers). The same evidence applies — 136,138 of 187,374 LOC are near-identical copies. The key additional patterns not covered by H4:

Frontend — 8 legacyFormatters files (covered in H2) contain identical function bodies across files; only the module-suffix string changes.

Frontend — 147 class-based JSX widgets (`resources/js/legacy/class/*ClassWidget*.jsx`) are 51-LOC components with identical `componentDidMount` fetch patterns and 35 hardcoded row-mirror divs.

`resources/js/legacy/class/WhisperCoachClassWidget2.jsx:3-15`

```
export default class WhisperCoachClassWidget2 extends React.Component {
  state = { count: 0, rows: [] }
  componentDidMount() {
    fetch('/ivr-legacy/whisper-coach/index').then(r => r.json())
      .then(d => this.setState({ rows: d.data || [] }))
  }
  render() {
    return (
      <div className="legacy-class-widget">
        <h3>WhisperCoach legacy class widget 2</h3>
        <button type="button" onClick={() => this.setState({ count:
this.state.count + 1 })}>
          {this.state.count}
        </button>
        // ... 35 identical row mirrors
```

Why it matters here: The sheer volume of duplication (72.6%) means the codebase's effective unique logic is only ~51,000 LOC. Maintenance cost, test coverage requirements, and bundle size are all inflated by ~3.6x for zero functional benefit.

Recommended approach:

1. Delete all duplicate frontend files after creating parameterized replacements.
2. Add an ESLint no-restricted-imports rule and a CI duplicate-detection step (e.g., `jscpd`) to prevent re-accumulation.

147 files affected.

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

<code>resources/js/legacy/class/TextToSpeechClassWidget2.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/TextToSpeechClassWidget3.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/TextToSpeechClassWidget4.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/TicketSyncClassWidget1.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/TicketSyncClassWidget3.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/TrunkGroupClassWidget1.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/TrunkGroupClassWidget3.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/TrunkGroupClassWidget4.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/VoicemailBoxClassWidget0.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/VoicemailBoxClassWidget1.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/VoicemailBoxClassWidget2.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/VoicemailBoxClassWidget4.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/WebhookDispatcherClassWidget0.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/WebhookDispatcherClassWidget2.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/WebhookDispatcherClassWidget4.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/WhisperCoachClassWidget0.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module
<code>resources/js/legacy/class/WhisperCoachClassWidget2.jsx</code>	—	147 identical class-based React widgets	Replace with single functional component parameterized by module

H9. Copy-Paste God Classes (additional) HIGH

Benchmark: Clone ratio = 100% (24/24 legacy service+repository classes are structural clones) → falls in the **High Risk** band (Good <5% · Moderate 5–20% · High Risk >20%).

All 12 GodService files and all 12 Repository files are verbatim copies with only the Eloquent model/table name swapped. Each GodService contains a hard-coded API key (`LEGACY_IVR_KEY_2022`), a mutable static cache (`$sharedRuntimeCache`), uses `extract($payload)` in every method (unsafe variable injection), and has a blocking `sleep(1)` call in every workflow method. The count of `extract()` calls across the codebase is **4,940** (controllers + services combined).

`app/Legacy/Services/AgentDeskGodService.php` vs `app/Legacy/Services/CallRoutingGodService.php` — diffing these files shows only the class name and table name differ across all 373 lines.

Why it matters here: The God Class anti-pattern (single class with 45 unrelated workflow methods, no interfaces, no dependency injection, mutable global state) makes the legacy layer untestable and resistant to any incremental improvement. The hard-coded

secrets and `extract()` usage are security concerns cross-cutting every module.

Recommended approach:

1. Extract a `LegacyWorkflowCommand` interface and implement one Command per workflow.
2. Inject dependencies (DB connection, cache) rather than using static state.
3. Remove all `extract()` calls; destructure explicitly.
4. Move secrets to environment configuration.

12 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Services/AgentDeskGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection
<code>app/Legacy/Services/CallFlowGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection
<code>app/Legacy/Services/CallRecordingGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection
<code>app/Legacy/Services/CallRoutingGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection
<code>app/Legacy/Services/CustomerProfileGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection
<code>app/Legacy/Services/DidInventoryGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection
<code>app/Legacy/Services/HistoricalReportsGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection
<code>app/Legacy/Services/LiveMonitoringGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection
<code>app/Legacy/Services/PromptLibraryGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection
<code>app/Legacy/Services/QueueManagementGodService.php</code>	8, 10, 15, 17, 23, 25, 31, 33, 39, 41, 47, 49 +80	12 identical God Classes with unsafe patterns	Replace with Command pattern and dependency injection

Not observed (rated Good): H1 (cyclomatic complexity — all functions are short, max ~6 branches), H3 (large functions — longest is ~21 LOC), H6 (churn — only 2 commits in 6 months), H7 (defect-prone files — only 1 fix commit found in history), H8 (ownership — clear single-author ownership per layer).

2.3 Code Churn & Stability Evidence

Git history is shallow — only 124 total commits across the repository's lifetime, with only **2 commits in the last 6 months** (both in July 2026, both from a single author "Shweta Shende" adding the IVR legacy layer).

Metric	Value
Total commits (all time)	124
Commits in last 6 months	2
Distinct authors (all time)	20
IVR layer authors	1 (Shweta Shende)
Core PingCRM top author	Jonathan Reinink (~80% of original commits)

Top files by churn (last 6 months):

File	Changes	Authors
README.md	2	1
routes/web.php	1	1
All IVR files	1 (initial commit)	1

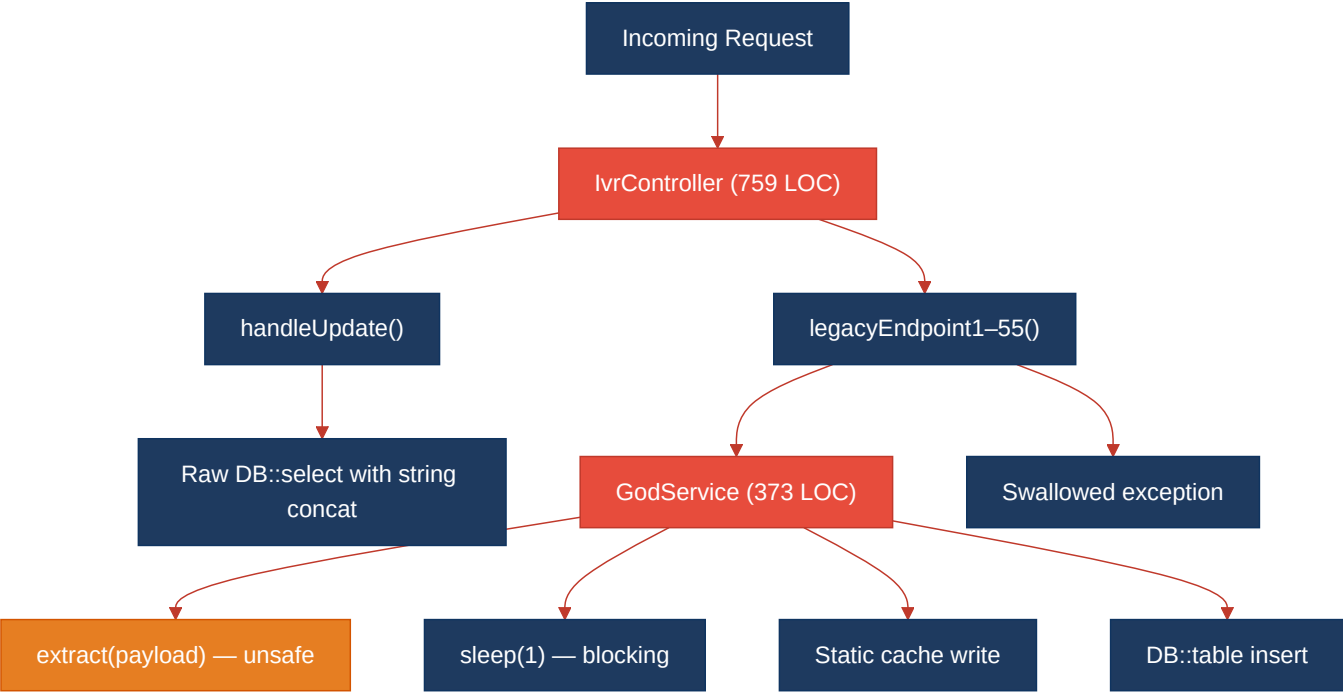
Defect-prone files (commits matching `fix|bug|hotfix` in message):

File	Fix commits
tests/Feature/ContactsTest.php	1
resources/views/app.blade.php	1
Legacy Vue/Webpack files (since removed)	1 each

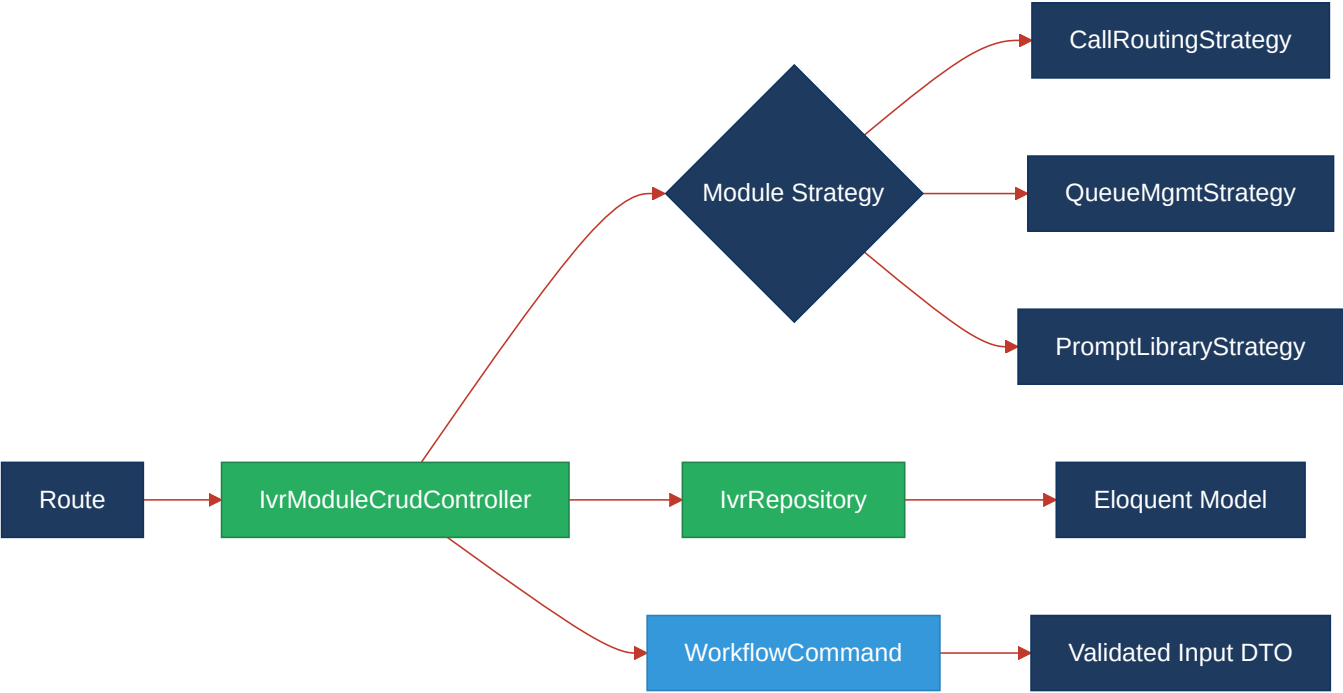
No file has more than 1 fix-related commit. The low churn reflects that the IVR layer was generated in a single batch, not iteratively developed.

2.4 Diagrams

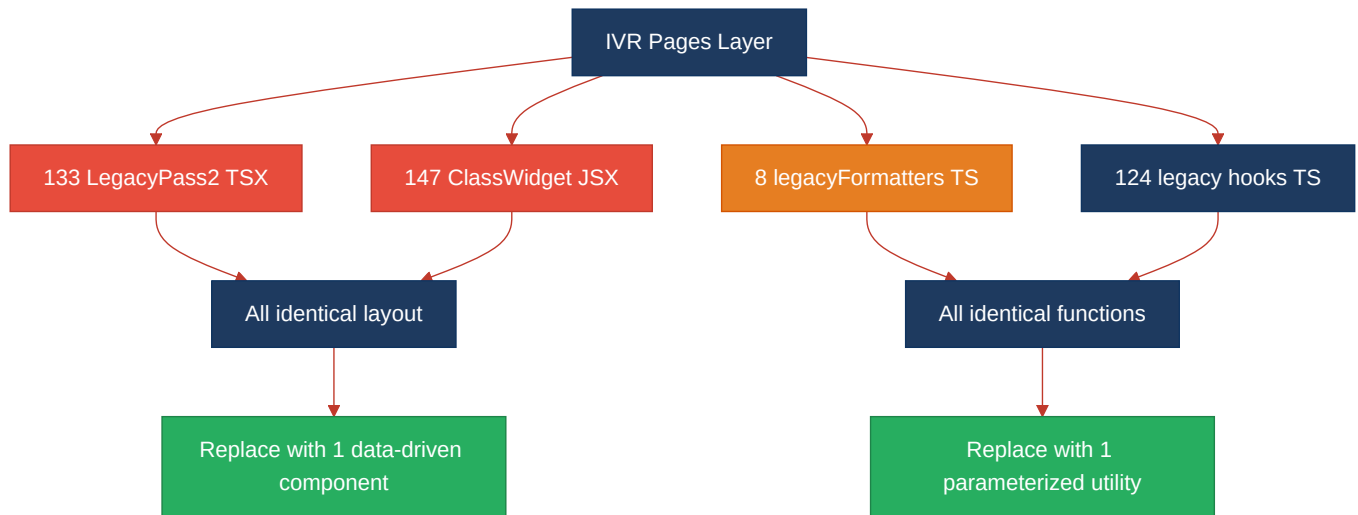
Complexity / call-flow hotspot



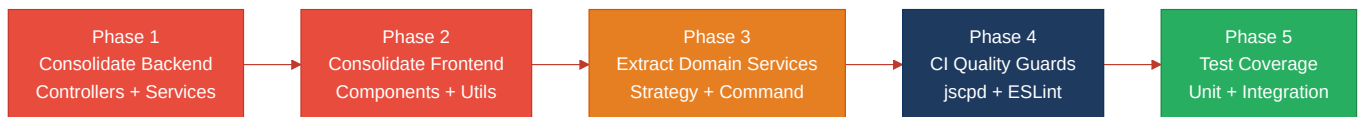
Refactored target structure



Frontend duplication structure



Improvement roadmap



2.5 Actions Required

Hotspot	Action	Rating	Priority
H2 — Large Classes	Replace 80 IVR controllers with 1 generic controller; consolidate 8 legacyFormatters into 1 utility	HIGH RISK	CRITICAL
H4 — Business Logic Duplication	Consolidate 12 GodServices into 1 parameterized service; consolidate 12 Repositories into 1 generic repository; replace 133 LegacyPass2 TSX with 1 data-driven component	HIGH RISK	CRITICAL
H5 — Duplicate Code (general)	Delete 147 class widgets after creating single functional replacement; add <code>jscpd</code> to CI; add ESLint no-restricted-imports rule	HIGH RISK	CRITICAL
H9 — Copy-Paste God Classes	Replace God Classes with Command pattern + DI; remove all <code>extract()</code> calls; move secrets to env config	HIGH RISK	HIGH

2.6 Expected Outcomes

- **~72% reduction in total LOC** by consolidating duplicate backend controllers, services, repositories, and frontend components into parameterized alternatives.
- **Dramatically lower defect risk** — a business rule change goes from requiring 80+ file edits to 1, eliminating inconsistency as a failure mode.
- **Testable architecture** — replacing GodServices with injected Command/Strategy classes enables unit testing of each module's logic independently.
- **Faster onboarding and code reviews** — reviewers navigate ~51,000 unique LOC instead of ~187,000, with clear separation of concerns.

- **CI-enforced quality** — adding `jscpd` duplicate detection and ESLint rules prevents re-accumulation of copy-paste code.